DreamPrice: A Causal DreamerV3 World Model for Offline Retail Pricing

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Abstract

Retail pricing is a high-stakes sequential decision problem in which every price change triggers a cascade of demand responses: consumers substitute across products, stockpile during promotions, and competitors react, with effects that propagate over weeks and quarters. Despite decades of work on demand estimation and price optimization, the systems used in practice rely on either structural econometric models with manually specified demand curves or model-free reinforcement learning methods that cannot perform counterfactual reasoning. No learned world model has been applied to economic environments where transition dynamics are endogenous to the agent's actions.

This paper introduces DreamPrice, to our knowledge the first Dreamer-style world model for retail pricing. The system combines DreamerV3's three-phase training recipe with a Mamba-2 selective state-space backbone, DRAMA-style decoupled posteriors, and a causally-constrained demand decoder. Causal price elasticities are pre-estimated via Hausman instrumental variables combined with double machine learning (DML-PLIV) and frozen into the decoder, preventing the model from learning confounded price-demand relationships from the observational data. The architecture replaces the gated recurrent unit (GRU) backbone of standard recurrent state-space models with Mamba-2's structured state-space duality (SSD), enabling $O(n)$ parallel training and $O(1)$-per-step recurrent imagination. MOPO-style pessimism via a 5-head reward ensemble guards against distributional shift inherent in offline learning.

Trained on the Dominick's Finer Foods scanner dataset (1989–1997, 93 stores, canned soup category with $\sim$25 stock-keeping units), DreamPrice achieves a world model ELBO loss of 22.44 and an actor return of 193.7 in the symlog-transformed reward space after 100K training steps. An ablation study across nine configurations confirms the contribution of each component: removing offline pessimism degrades return by 85.6%, removing symlog and twohot representations increases world model loss by 25$\times$, and the Mamba-2 backbone yields 28.9% higher return than a GRU alternative.

1 Introduction

Retail pricing ranks among the most consequential sequential decision problems in operations research and marketing. Every price change triggers a cascade of demand responses: consumers substitute across stock-keeping units (SKUs), stockpile during promotions, and competitors react. These dynamics unfold over weeks and quarters, with effects that propagate through substitution networks and depend on store demographics, seasonality, and latent competitive context. Despite decades of empirical work on price elasticity and demand estimation [Hoch et al., 1995, Nevo, 2001, Montgomery, 1997], the systems that support actual pricing decisions today fall into two camps. Structural econometric models, which combine parametric demand curves with mathematical optimization (linear or quadratic programming, Ramsey pricing), encode assumptions about demand and competition, but their fidelity to real-world dynamics is

Table 1: Taxonomy of sequential decision environments by domain and dynamics model. DreamPrice is, to our knowledge, the first Dreamer-style world model trained directly on retail scanner data.

	Learned model	**Structural / rule-based model**
Physical	DreamerV1–V3 [Hafner et al., 2020, 2021, 2023], IRIS [Micheli et al., 2023], TD-MPC2 [Hansen et al., 2024]	MuJoCo, Isaac Gym, CARLA
Economic	**DreamPrice (this work)**	ABIDES [Byrd et al., 2020], DSGE [Smets and Wouters, 2007], AI Economist [Zheng et al., 2020]

limited by the tractability of closed-form solutions and the static, single-period nature of the optimization. Model-free reinforcement learning methods [Chen and Simchi-Levi, 2022, Ban and Keskin, 2021] learn policies directly from data but forgo an explicit dynamics model, making counterfactual reasoning and imagination-based planning impossible. No learned world model exists for economic environments.[1]

This gap can be formalized using a simple taxonomy. Consider the space of sequential decision environments along two axes: the domain (physical versus economic) and the nature of the dynamics model (learned versus structural or rule-based). In the physical domain, learned world models have achieved considerable success. The Dreamer family [Hafner et al., 2020, 2021, 2023] pioneered recurrent state-space models (RSSMs) for latent imagination and policy optimization. MuZero [Schrittwieser et al., 2020] demonstrated that planning with a learned model suffices for mastering board games and Atari. More recently, IRIS [Micheli et al., 2023] and TD-MPC2 [Hansen et al., 2024] have extended these ideas to continuous control and high-dimensional observations. These methods rely on engineered physical simulators such as MuJoCo, Isaac Gym, or CARLA for evaluation and transfer. In the economic domain, by contrast, the landscape is dominated by structural models: agent-based simulators like ABIDES, parametric demand models in the spirit of Berry et al. [1995] and Nevo [2001], and the AI Economist [Zheng et al., 2020], which uses a rule-based economic environment for tax policy experiments. The quadrant of learned models applied to economic domains has remained empty. DreamPrice fills this gap.

Why do world models matter for pricing? Imagination-based planning enables counterfactual reasoning at scale. A retailer can ask: what if the price of SKU 3 were cut by 5% next quarter? A learned world model rolls out latent trajectories under that hypothetical action, predicting demand responses, substitution effects, and downstream margin impact without executing the change in the real world. Such queries are infeasible with model-free methods, which require either costly online exploration or restrictive assumptions about the policy class. Offline learning from historical scanner data [James M. Kilts Center for Marketing, 1999] further aligns with the constraints of retail: experimentation is expensive, and the Dominick's dataset alone spans 400 weeks across 93 stores and roughly 18,000 universal product codes (UPCs). The structured economic state space (prices, quantities, demographics, and derived features in 100–300 dimensions) is fundamentally more tractable than the 64×64×3 image observations that DreamerV3 was designed for [Hafner et al., 2023]. A world model trained on such tabular data can capture demand dynamics, promotional effects, and competitive structure without the burden of high-dimensional visual encoding.

Economic data, however, poses a challenge absent in physical domains: endogeneity. In

[1] We use "economic environments" to mean retail or consumer markets with scanner-data observations and endogenous pricing dynamics. Learned simulators for synthetic market-making or limit-order-book reinforcement learning operate on fundamentally different state spaces (continuous order flow, millisecond time scales, and exogenous price processes) and do not address the demand-estimation and causal-identification challenges central to retail pricing.

physical simulators, the transition dynamics are exogenous to the agent; the laws of physics do not depend on the policy. In retail, prices and demand are simultaneously determined. Retailers set prices in response to expected demand, costs, and competitive pressure; demand, in turn, responds to those prices. Naive machine learning that regresses demand on price learns a confounded relationship, conflating the causal effect of price with selection and omitted variables [Hausman, 1978]. The resulting elasticity estimates are biased, and a world model built on such estimates would produce misleading counterfactuals. The econometric literature has long addressed this through instrumental variables [Hausman, 1978], control functions [Berry et al., 1995], and more recently double machine learning [Chernozhukov et al., 2018, Bach et al., 2022]. DreamPrice integrates these identification strategies by pre-estimating causal price elasticities via DML-PLIV with Hausman-style instruments (described in Section 3) and freezing them into a constrained demand decoder. The decoder's residual MLP learns the remainder, including seasonality, demographics, and promotion effects, while the price-demand relationship remains causally identified.

The contributions of this paper are as follows:

1. **A learned world model for economic environments.** To our knowledge, DreamPrice is the first Dreamer-style world model trained directly on retail scanner data, enabling imagination-based planning and counterfactual demand estimation in a domain previously served only by structural simulators or model-free methods.

2. **Mamba-2 SSM backbone.** The recurrent backbone of standard RSSMs is replaced with Mamba-2 [Dao and Gu, 2024], a selective state-space model that achieves $O(n)$ parallel training via structured state-space duality (SSD) and $O(1)$-per-step recurrent rollouts during imagination. This replaces the GRU-based recurrence of DreamerV3 [Hafner et al., 2023] with a more scalable alternative grounded in the S4 [Gu et al., 2022] and Mamba [Gu and Dao, 2024] lineage.

3. **DRAMA-style decoupled posterior.** Following Rajbhandari et al. [2024], the posterior over the stochastic latent depends only on the current observation, not on the recurrent hidden state. This decoupling breaks the sequential dependency that would otherwise prevent Mamba-2's parallel SSD scan during training. During imagination, the model switches to recurrent `step()` mode for single-step rollouts.

4. **Causally-constrained demand decoder.** Per-category price elasticities are pre-estimated via DML-PLIV [Chernozhukov et al., 2018] and frozen into the DreamPrice decoder. The decoder enforces the causal price-demand relationship while an MLP residual learns seasonality, demographics, and promotion effects. This prevents the world model from learning confounded elasticities from the observational data.

5. **MOPO-style offline pessimism.** To guard against distributional shift inherent in offline reinforcement learning [Levine et al., 2020], DreamPrice employs a 5-head reward ensemble and uses the lower confidence bound $r_{\text{pessimistic}} = r_{\text{mean}} - \lambda_{\text{LCB}} \cdot r_{\text{std}}$ for policy optimization [Yu et al., 2020]. The actor is trained on this pessimistic reward signal while the critic and world model remain unchanged.

The remainder of this paper is organized as follows. Section 2 reviews related work on world models, state-space models, causal demand estimation, and offline reinforcement learning. Section 3 presents the DreamPrice architecture, training procedure, and causal identification strategy. Section 4 reports experiments on the Dominick's canned soup category, including world model quality, elasticity recovery, and offline policy performance. Section 5 concludes and discusses limitations and future directions.

2 Related Work

This section situates DreamPrice within seven strands of literature: traditional demand forecasting and price optimization, learned world models for sequential decision making, structured state-space models, causal inference in demand estimation, offline reinforcement learning, pricing optimization and retail analytics, and the Dominick's scanner dataset as the canonical empirical setting.

2.1 Traditional Demand Forecasting and Price Optimization

The standard approach to retail pricing proceeds in two stages: estimate demand as a function of price and other covariates, then optimize prices subject to business constraints. On the forecasting side, classical time-series methods, including autoregressive integrated moving average (ARIMA) models, exponential smoothing, and seasonal decomposition, have long served as workhorses for retail demand prediction [Fildes et al., 2022]. These methods model temporal patterns in aggregate sales but do not capture cross-product substitution or the causal effect of price changes on demand. Machine learning methods, including gradient boosting machines [Friedman, 2001], random forests, and neural networks, have improved point forecast accuracy by incorporating high-dimensional features such as promotions, holidays, competitor prices, and weather. Fildes et al. [2022] provide a comprehensive survey of retail forecasting methods, documenting that ML approaches now dominate industry practice for point forecasting while statistical methods remain competitive for uncertainty quantification.

On the optimization side, the dominant paradigm is to estimate price elasticities, typically via log-linear demand models or the Berry-Levinsohn-Pakes (BLP) framework [Berry et al., 1995], and then solve a constrained optimization problem. Common formulations include linear or quadratic programming to maximize gross margin subject to price-range constraints, competitive parity rules, and promotional calendars [Talluri and van Ryzin, 2004]. Ramsey pricing, which sets markups inversely proportional to demand elasticity, provides a welfare-theoretic benchmark. These two-stage approaches are transparent and interpretable but fundamentally limited: the demand model is static (fit once, applied repeatedly), the optimization horizon is typically a single period, and the system cannot capture multi-step dynamics such as stockpiling, reference-price effects, or competitive responses that unfold over weeks. DreamPrice replaces both stages with an end-to-end learned dynamics model that supports multi-step imagination and policy optimization over arbitrary horizons.

2.2 World Models for Sequential Decision Making

The idea of learning a world model, a compact representation of environment dynamics that supports prediction and planning, dates to Ha and Schmidhuber [2018], who demonstrated that a variational autoencoder could learn a latent space in which an agent could imagine future trajectories. That foundational work showed that an agent could learn to navigate a maze by training purely in imagined latent space, without ever observing the true environment during policy learning. The subsequent decade has seen a steady progression of architectures that refine this idea.

The Dreamer family [Hafner et al., 2020, 2021, 2023] established the recurrent state-space model (RSSM) as the dominant paradigm for learning world models. DreamerV1 [Hafner et al., 2020] introduced latent imagination: the agent learns a stochastic latent representation that evolves over time, with a decoder predicting observations and rewards from the latent state. DreamerV2 [Hafner et al., 2021] extended this to discrete action spaces by mastering Atari games through imagination-based policy optimization, demonstrating that world models could scale to high-dimensional pixel observations. DreamerV3 [Hafner et al., 2023] unified the design across discrete and continuous domains, introducing symlog normalization, twohot distributions

for rewards and values, and KL balancing to prevent posterior collapse. The RSSM architecture, a deterministic recurrent core (traditionally a GRU) augmented with stochastic latent variables, has become the standard for model-based reinforcement learning in physical environments.

Parallel developments have explored alternative architectures. MuZero [Schrittwieser et al., 2020] demonstrated that planning with a learned model suffices for mastering board games and Atari, using a learned dynamics model that predicts value and policy targets rather than raw observations. The model is trained end-to-end with Monte Carlo tree search, and the resulting system achieves superhuman performance in Go, chess, and shogi. IRIS [Micheli et al., 2023] and TD-MPC2 [Hansen et al., 2024] have extended world models to continuous control and high-dimensional observations. IRIS replaces the RSSM with a transformer-based architecture, using discrete tokens for the latent representation and achieving strong performance on DMC and Atari. TD-MPC2 combines model-based planning with temporal difference learning for continuous control, scaling to complex robotics tasks.

A critical observation unifies these works: they all operate on physical environments. The Dreamer family, MuZero, IRIS, and TD-MPC2 are evaluated on games (Atari, board games), physical simulators (MuJoCo, Isaac Gym), or robotic control. The dynamics they learn are governed by physics or game rules: the transition from state to state is exogenous to the agent's policy in the sense that the laws of motion do not change with the agent's behavior. In economic environments, by contrast, the transition dynamics are endogenous. Prices and demand are simultaneously determined; retail prices respond to expected demand, costs, and competitive pressure, while demand responds to those prices. No learned world model has been applied to such economic domains. DreamPrice fills this gap by learning the first retail dynamics model from scanner data.

2.3 Structured State Space Models

The recurrent backbone of standard RSSMs is typically a gated recurrent unit (GRU) or long short-term memory (LSTM) network. These architectures process sequences sequentially, with each time step depending on the previous hidden state. For long sequences, this sequential dependency can be a bottleneck; the GRU cannot exploit parallel computation across the time dimension during training.

Structured state space models (SSMs) offer an alternative that combines long-range dependency modeling with computational efficiency. Gu et al. [2022] introduced S4 (Structured State Spaces), which parameterizes sequences using linear state-space equations with learned transition matrices. S4 achieves strong performance on long-range dependencies, including image classification and audio generation, by using the HiPPO initialization for the state matrix and efficient parallel scan algorithms. The key insight is that linear recurrence admits a closed-form solution that can be computed in parallel via associative scan operations.

Mamba [Gu and Dao, 2024] extended S4 with selective gating: the state transition parameters become input-dependent rather than fixed. This selectivity allows the model to focus on relevant information at each step, analogous to the gating in attention mechanisms. Mamba achieves linear-time complexity in sequence length and scales to long contexts without the quadratic cost of attention. Mamba-2 [Dao and Gu, 2024] further unifies these ideas through structured state-space duality (SSD), which establishes a formal connection between linear attention and state-space models. The SSD framework shows that both can be viewed as instances of a more general class of models, enabling algorithms that share the best properties of each: parallel training with $O(n)$ complexity in sequence length, and recurrent inference with $O(1)$ cost per step.

This duality has direct implications for world models. During training, the model processes sequences of observations and actions in parallel: the entire batch of sequences can be scanned in one pass, as the posterior over the stochastic latent depends only on the current observation in the DRAMA-style decoupled design [Rajbhandari et al., 2024]. During imagination, the

model switches to recurrent mode: the Mamba-2 block's `step()` function computes the next hidden state from the previous state and the current input in constant time per step. This matches the training-then-imagination workflow of DreamerV3, but with a backbone that scales more efficiently to long sequences and avoids the sequential bottleneck of GRU recurrence. DreamPrice adopts this architecture, replacing the GRU with Mamba-2 for the recurrent core.

2.4 Causal Inference in Demand Estimation

Estimating demand as a function of price is a central problem in empirical industrial organization and marketing. The naive approach of regressing quantity on price fails because prices are endogenous. Retailers set prices in response to expected demand, costs, and competitive conditions; unobserved factors that correlate with both price and demand (e.g., product quality, seasonality, local shocks) confound the regression. The resulting elasticity estimates are biased, and a world model built on such estimates would produce misleading counterfactuals.

The econometric literature has developed several identification strategies. The Berry-Levinsohn-Pakes (BLP) framework [Berry et al., 1995] uses a structural model of discrete choice with random coefficients; identification relies on moment conditions that match predicted choice shares to observed shares. Nevo [2001] applied this approach to the ready-to-eat cereal industry, estimating elasticities and measuring market power. Hausman [1978] introduced the use of instrumental variables for specification testing and parameter identification. When prices are correlated across stores or time for reasons exogenous to store-specific demand shocks, cross-sectional or temporal variation can serve as instruments. The Hausman IV approach, which uses the average log price in other stores as an instrument for a given store's price, exploits the panel structure of scanner data: prices in other stores are relevant (drivers of cost and competition are correlated) but plausibly exogenous to store-specific demand shocks.

Hoch et al. [1995] applied this logic to the Dominick's dataset, estimating store-level price elasticities using cross-store instruments. Their work demonstrated that elasticity varies substantially across stores and product categories, with important implications for targeted pricing. Park and Gupta [2010] proposed a nonparametric copula-based approach to demand estimation with endogeneity, using the joint distribution of price and control variables to construct a control function. The copula correction does not require specifying instruments ex ante but relies on distributional assumptions.

The double machine learning (DML) framework of Chernozhukov et al. [2018] provides a flexible approach that can incorporate any of these identification strategies. DML partitions the estimation into orthogonal nuisance parameters and parameters of interest, using cross-fitting to avoid overfitting bias. The framework accommodates high-dimensional controls and nonparametric first-stage estimators. Bach et al. [2022] provide a Python implementation that supports DML-PLIV (partial linear instrumental variables) for demand estimation with endogenous prices.

The key contribution of DreamPrice in this literature is the integration of econometric identification into a neural world model. Rather than estimating demand entirely from data, which would conflate the causal effect of price with confounders, DreamPrice pre-estimates causal price elasticities via DML-PLIV and freezes them into the demand decoder. The decoder's residual MLP learns the remainder, including seasonality, demographics, promotion effects, and substitution patterns, while the price-demand relationship remains causally identified. Freezing econometric estimates into neural network decoders is novel; prior work in demand estimation either uses parametric structural models or applies machine learning without causal constraints.

2.5 Offline Reinforcement Learning

Reinforcement learning methods can be classified along two orthogonal axes: the data regime (online versus offline) and the use of a dynamics model (model-free versus model-based). Online

methods interact with the environment during training, collecting new experience and refining their policy through exploration. Model-free online methods, such as DQN [Mnih et al., 2015], PPO [Schulman et al., 2017], and SAC [Haarnoja et al., 2018], learn value functions or policies directly from transitions without building an explicit dynamics model. Model-based online methods, including the Dreamer family [Hafner et al., 2023] and MuZero [Schrittwieser et al., 2020], learn a dynamics model and optimize the policy in imagination, reducing the number of real-world interactions required.

In many settings, retail pricing among them, online exploration is infeasible or prohibited: experimenting with prices on real customers carries financial risk and ethical constraints. The agent must learn from a fixed dataset of historical trajectories collected under some (possibly unknown) behavioral policy. This offline learning setting introduces the distribution shift problem: the policy being optimized may visit states and actions that differ from the data distribution, and the learned value function or dynamics model may extrapolate poorly outside the support of the data [Levine et al., 2020].

Model-free offline methods address distribution shift through value-function conservatism or policy constraints. Batch-Constrained Q-learning (BCQ) [Fujimoto et al., 2019] restricts the policy to actions observed in the dataset by using a generative model of the behavioral policy. Conservative Q-Learning (CQL) [Kumar et al., 2020] penalizes the Q-function for assigning high values to out-of-distribution actions, effectively learning a lower bound on the true Q-function. Implicit Q-Learning (IQL) [Kostrikov et al., 2022] avoids querying out-of-distribution actions entirely by using expectile regression to approximate the optimal value function from the dataset. Decision Transformer [Chen et al., 2021] reframes offline RL as a sequence prediction problem, conditioning a transformer on desired returns to generate actions. These methods learn policies but cannot perform counterfactual reasoning or multi-step imagination.

Model-based offline methods combine a learned dynamics model with pessimism to handle distribution shift. MOReL [Kidambi et al., 2020] constructs a pessimistic uncertainty set around the learned dynamics model and restricts planning to states that are provably within the uncertainty set. MOPO [Yu et al., 2020] takes a softer approach: it modifies the reward signal during imagination rather than constraining the policy or dynamics. The key idea is that the learned dynamics model is uncertain in regions of state-action space that are poorly covered by the data. MOPO uses an ensemble of dynamics models to estimate this uncertainty; the reward is penalized by the uncertainty, so the agent is discouraged from visiting states where the model is unreliable. The penalty is $r_{\text{pessimistic}} = r_{\text{mean}} - \lambda_{\text{LCB}} \cdot r_{\text{std}}$, where r_{mean} and r_{std} come from the ensemble. COMBO [Yu et al., 2021] combines model-based and conservative approaches, using offline data to augment the model rollout and applying conservative regularization to the value function.

DreamPrice occupies the model-based offline quadrant: it learns a world model from historical scanner data and trains a policy entirely in imagination, using MOPO-style pessimism to guard against distributional shift. The MOPO approach is particularly well-suited to the DreamerV3 training loop. DreamerV3 already trains an actor in imagination; the only modification is to change the reward signal used during actor training. The critic and world model remain unchanged. A 5-head reward ensemble provides r_{mean} and r_{std}; the actor is trained on the pessimistic reward while the critic continues to predict the true return. This minimal modification preserves the three-phase structure and avoids the complexity of full conservative value estimation.

2.6 Pricing Optimization and Retail Analytics

The application of reinforcement learning to pricing has a substantial literature. Chen and Simchi-Levi [2022] study dynamic pricing with reinforcement learning, formulating the problem as a Markov decision process and analyzing the trade-off between exploration and exploitation. Ban and Keskin [2021] extend this to personalized pricing with heterogeneous elasticity, using

machine learning to estimate demand as a function of high-dimensional features. The AI Economist [Zheng et al., 2020] applies multi-agent reinforcement learning to tax policy design, training agents in a rule-based economic environment to learn tax schedules that balance equality and productivity.

These works share a common limitation: they rely on structural simulators or model-free methods. Chen and Simchi-Levi assume a known or estimated demand model; Ban and Keskin use model-free RL with feature-based demand estimation. The AI Economist uses a rule-based economic environment for training. None of these approaches learns a dynamics model from data. The gap is twofold: (1) no learned world model exists for pricing environments, and (2) no imagination-based planning has been applied to retail. A model-free agent can learn a policy from historical data, but it cannot answer counterfactual queries such as "what if the price of SKU 3 were cut by 5% next quarter?" without executing the change. A learned world model enables such queries by rolling out latent trajectories under hypothetical actions.

DreamPrice is the first system to learn a dynamics model for retail pricing from scanner data. The model captures demand responses to prices, promotions, and seasonality; it supports imagination-based policy optimization; and it integrates causal identification to prevent confounded elasticity estimates. The architecture is designed for the structured economic state space (prices, quantities, demographics, and derived features in 100–300 dimensions), which is fundamentally more tractable than the high-dimensional pixel observations that DreamerV3 was designed for.

2.7 Scanner Data and the Dominick's Dataset

The Dominick's Finer Foods dataset [James M. Kilts Center for Marketing, 1999] has become the canonical dataset for empirical retail and marketing research. Maintained by the James M. Kilts Center for Marketing at the University of Chicago Booth School of Business, the dataset spans 400 weeks (September 1989 to May 1997), 93 stores, and approximately 18,000 unique UPCs across 29 product categories. The movement files record store-SKU-week tuples with unit sales, prices, profit margins, and promotion indicators. The data structure enables rich panel analysis: the same products are observed across stores and over time, with variation in prices driven by local competition, cost shocks, and promotional calendars.

Hoch et al. [1995] used the Dominick's data to estimate store-level price elasticities, demonstrating that elasticity varies substantially across stores and that store demographics (income, education, competition) explain much of this variation. Montgomery [1997] estimated heterogeneous price thresholds, the price levels at which consumers switch between brands, using the same dataset. The panel structure (93 stores) supports Hausman-style instruments: the average log price in other stores for a given UPC-week serves as an instrument for a store's own price, as it is correlated with cost and competition but plausibly exogenous to store-specific demand shocks. The store demographic file, with 300+ variables from the 1990 Census, enables control for heterogeneity in consumer preferences and competitive intensity.

The Dominick's dataset is ideal for training a learned world model. It directly contains state-action-outcome tuples: the action is the price and promotion decision, the state includes competitor context, seasonality, and store demographics, and the outcome is observed demand. The PROFIT field enables wholesale cost derivation. The temporal split (train weeks 1–280, validation 281–340, test 341–400) is strictly chronological, avoiding look-ahead bias. Categories such as canned soup, beer, and soft drinks offer moderate SKU counts with clear substitution patterns and promotional activity. DreamPrice is trained on this dataset, with preprocessing that constructs Hausman IVs, computes derived features, and applies symlog normalization for the world model encoder.

3 Methodology

This section presents the DreamPrice architecture, training procedure, and causal identification strategy. The approach combines a recurrent state-space model with a Mamba-2 selective state-space backbone [Dao and Gu, 2024], DRAMA-style decoupled posteriors [Rajbhandari et al., 2024], a causally-constrained demand decoder, and MOPO-style offline pessimism [Yu et al., 2020]. The training recipe follows DreamerV3 [Hafner et al., 2023] with adaptations for retail pricing and offline learning.

3.1 Problem Formulation

Retail pricing is formulated as a partially observable Markov decision process (POMDP) in which the decision-maker observes prices, quantities sold, promotions, and store demographics, but the underlying state governing demand dynamics is only partially observable. The true state s_t comprises demand state (consumer preferences, inventory effects, substitution structure), competitor state (rival prices and promotions, which are unobserved in the Dominick's dataset), and seasonal state (calendar effects, holidays, macroeconomic conditions). The observation x_t at time t includes per-SKU prices, units sold, promotion indicators, lagged demand and price features, rolling statistics, and store demographics. The action a_t is a vector of price multipliers over K SKUs, discretized to 21 levels per SKU: $a_t \in \{0.90, 0.91, \ldots, 1.10\}^K$, representing percentage adjustments relative to a baseline price. The reward combines gross margin with a volatility penalty and a margin floor penalty to discourage degenerate strategies such as perpetual deep discounts or excessive price churn. The transition dynamics are unknown and must be learned from data.

Definition 1 (Retail Pricing POMDP)**.** *A retail pricing POMDP is a tuple* $(\mathcal{S}, \mathcal{X}, \mathcal{A}, \mathcal{T}, \mathcal{O}, \mathcal{R}, \gamma)$ *where:*

- $\mathcal{S}$ *is the latent state space comprising demand, competitor, and seasonal components.*
- $\mathcal{X}$ *is the observation space: per-SKU prices, units sold, promotions, demographics, and derived features.*
- $\mathcal{A} = \{0.90, 0.91, \ldots, 1.10\}^K$ *is the action space of discrete price multipliers.*
- $\mathcal{T}(s_{t+1} \mid s_t, a_t)$ *is the unknown transition kernel.*
- $\mathcal{O}(x_t \mid s_t)$ *is the observation model.*
- $\mathcal{R}(s_t, a_t)$ *is the reward: gross margin minus volatility penalty minus margin floor penalty.*
- $\gamma \in (0, 1)$ *is the discount factor.*

The objective is to learn a world model that approximates $\mathcal{T}$ and $\mathcal{O}$ from offline historical data, then use imagination-based policy optimization to find a pricing policy that maximizes expected cumulative discounted reward [Sutton and Barto, 2018, Hafner et al., 2023].

3.2 RSSM Architecture

The recurrent state-space model (RSSM) at the core of DreamPrice consists of two components: a deterministic recurrent state h_t and a stochastic latent z_t. The deterministic state h_t is produced by the backbone (a Mamba-2 selective state-space model that replaces the GRU of standard DreamerV3 [Hafner et al., 2023]). The stochastic latent z_t follows a categorical distribution over 32 independent categorical variables, each with 32 classes, yielding a latent dimension of 1024. Inspired by the decoupling principle in DRAMA [Rajbhandari et al., 2024], the posterior

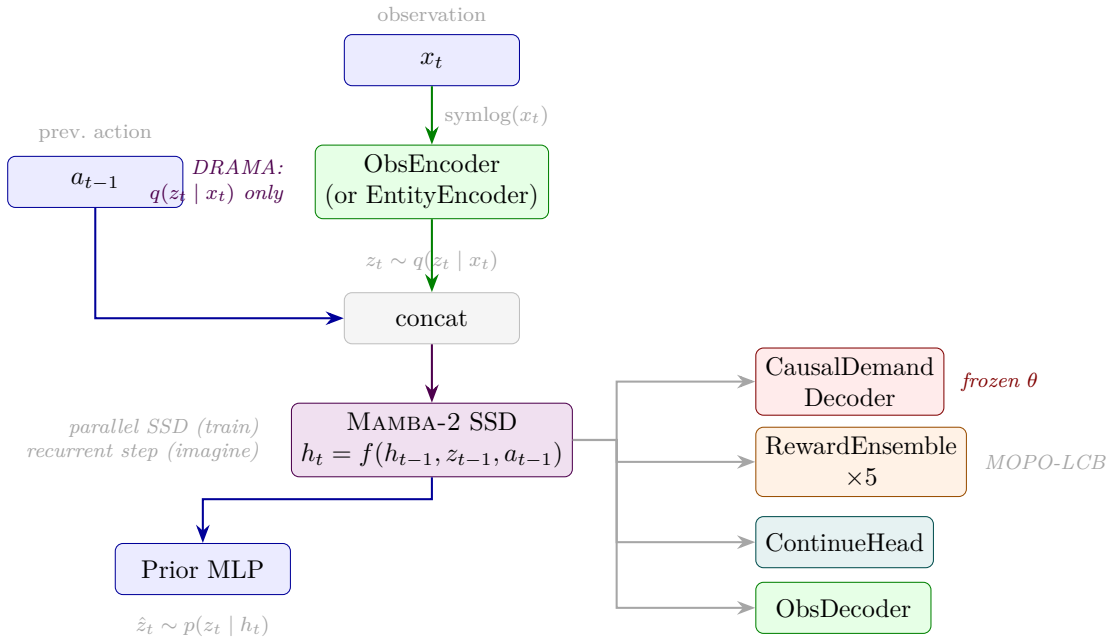

Figure 1: Overview of the DreamPrice architecture. The encoder produces the posterior z_t from the observation alone (DRAMA-style decoupled posterior). The Mamba-2 backbone processes sequences in parallel during training (SSD scan) and recurrently during imagination. The causal demand decoder freezes DML-PLIV elasticities; the 5-head reward ensemble provides MOPO-LCB pessimism.

$q(z_t \mid x_t)$ is computed from the observation alone via the encoder, without conditioning on h_t, which enables parallel training. The prior $p(z_t \mid h_t)$ is computed from the backbone output h_t via an MLP head.

Formally, the model is specified as follows. The backbone receives the concatenation of the previous latent z_{t-1} and the previous action a_{t-1}, and produces the deterministic state:

$$h_t = f_{\text{backbone}}(h_{t-1}, z_{t-1}, a_{t-1}) \tag{1}$$

where the backbone uses the Mamba-2 SSD.[2] The posterior over the stochastic latent depends only on the current observation:

$$z_t \sim q(z_t \mid x_t) = \text{Categorical}(32 \times 32, \text{straight-through} + \text{unimix}) \tag{2}$$

The prior over the next latent is predicted from the backbone state:

$$\hat{z}_t \sim p(z_t \mid h_t) = \text{MLP}(h_t) \tag{3}$$

The latent z_t uses 32 independent categorical distributions with 32 classes each, giving a total latent dimension of 1024. Straight-through gradients [Jang et al., 2017] are used: in the forward pass, a one-hot sample is drawn; in the backward pass, gradients flow through the soft probabilities. A uniform mixing (unimix) of 1% is applied to all categorical distributions: $\text{probs} = 0.99 \cdot \text{softmax}(\text{logits}) + 0.01/32$. This prevents near-deterministic distributions that destabilize KL computation and encourages exploration [Hafner et al., 2023].

[2] In the Mamba-2 implementation, the SSM state is maintained implicitly by the inference cache rather than passed as an explicit hidden vector. The mathematical notation uses h_t for clarity.

3.3 Mamba-2 Backbone

The standard RSSM backbone in DreamerV3 is a gated recurrent unit (GRU). DreamPrice replaces this with Mamba-2's structured state-space duality (SSD) [Dao and Gu, 2024, Gu and Dao, 2024]. The SSD formulation enables two computational modes. During training, the full sequence is processed in parallel via a scan over the sequence length, yielding $O(n)$ complexity in sequence length n. During imagination, the model switches to recurrent `step()` mode, producing one output per step with $O(1)$ cost per step in sequence length. This is critical for efficient rollout during actor training: the imagination horizon is 13 steps (one retail quarter), and recurrent inference avoids the memory and compute cost of a full parallel forward pass.

The backbone input at each step is the concatenation of the previous latent z_{t-1} and the previous action a_{t-1}, projected to dimension d_{model} by a linear layer. The output h_t has dimension d_{model} (typically 512). The internal SSM state is managed internally by the Mamba-2 library; during inference, `InferenceParams` maintains the state and is reset at episode boundaries.

The structured state-space duality expresses the recurrence as a linear recurrence with content-dependent gating. For input $u_t \in \mathbb{R}^{d_{\text{model}}}$, the SSM computes:

$$h_t = \bar{A}_t h_{t-1} + \bar{B}_t u_t, \quad y_t = C_h h_t \tag{4}$$

where $\bar{A}_t, \bar{B}_t$ are derived from discretized SSM parameters that depend on the input u_t via selective gating. The full derivation is given by Dao and Gu [2024]. When CUDA is unavailable or tensors are on CPU, the backbone transparently falls back to a GRU so that training and evaluation can proceed without the Mamba-2 library.

3.4 Decoupled Posterior (DRAMA-style)

In the standard DreamerV3 architecture, the posterior over the stochastic latent depends on both the observation and the recurrent state: $q(z_t \mid x_t, h_t)$ [Hafner et al., 2023]. This creates a sequential dependency: to compute z_t, one must first compute h_t, which depends on z_{t-1} and the backbone recurrence. As a consequence, the backbone cannot be run in parallel over the full sequence during training, because each h_t depends on the preceding h_{t-1} and z_{t-1}.

Extending the DRAMA decoupling idea [Rajbhandari et al., 2024] to the RSSM posterior, we have $q(z_t \mid x_t)$ only. The posterior now depends solely on the observation x_t, which is available for all t in the training batch. The backbone can compute h_t for all t in parallel, given the sequence of z_t and a_t (which are computed in a first pass from the encoder). The prior $p(z_t \mid h_t)$ still depends on h_t, so the KL divergence between posterior and prior provides a learning signal for the backbone. The evidence lower bound (ELBO) derivation remains valid: the objective is

$$\begin{aligned}\mathcal{L} = \mathbb{E}_{q(z_{1:T}|x_{1:T})}\Big[\sum_{t=1}^{T}(&\log p(x_t \mid z_t) + \log p(r_t \mid h_t) \\ &- \beta_{\text{dyn}}\text{KL}[\text{sg}(q(z_t \mid x_t))\|p(z_t \mid h_t)] - \beta_{\text{rep}}\text{KL}[q(z_t \mid x_t)\|\,\text{sg}(p(z_t \mid h_t))])\Big]\end{aligned} \tag{5}$$

where sg denotes the stop-gradient operator. The dynamics loss trains the prior (and thus the backbone) by matching $\text{sg}(q)$ to p; the representation loss trains the encoder by matching q to $\text{sg}(p)$. The KL terms still provide gradients to the backbone because $p(z_t \mid h_t)$ depends on h_t, and the backbone produces h_t.

3.5 Observation Encoder and Decoder

The observation encoder maps the raw observation x_t to a latent representation used by the posterior. The `ObsEncoder` is a 2-layer MLP with SiLU activations and RMSNorm: Linear(obs_dim $\to d_{\text{model}}$) $\to$ SiLU $\to$ RMSNorm $\to$ Linear($d_{\text{model}} \to d_{\text{model}}$) $\to$ SiLU $\to$

RMSNorm. The input to the encoder is $\text{symlog}(x_t)$; the symlog transform $\text{symlog}(x) = \text{sign}(x) \cdot \ln(|x| + 1)$ is applied to all continuous features (prices, quantities, demographics) so that the scale is appropriate for neural network training and the transform is defined at zero [Hafner et al., 2023].

For entity-factored variants, the `EntityEncoder` produces per-entity embeddings. Each entity (e.g., SKU-store) has embeddings for UPC, store, brand, and month, plus continuous features (price, demand, promotion depth, etc.). These are concatenated and passed through a fusion MLP to produce a d_{model}-dimensional output per entity.

The observation decoder reconstructs x_t from the concatenation of h_t and z_t. It is a 3-layer MLP: $\text{Linear}(d_{\text{model}} + \text{latent_dim} \to d_{\text{model}}) \to \text{SiLU} \to \text{Linear}(d_{\text{model}} \to d_{\text{model}}) \to \text{SiLU} \to \text{Linear}(d_{\text{model}} \to \text{obs_dim})$. The reconstruction target is $\text{symlog}(x_t)$, and the loss is the mean squared error between the predicted and target values. This provides the reconstruction term in the ELBO [Kingma and Welling, 2014].

3.6 Causal Demand Decoder

A central challenge in retail pricing is endogeneity: prices and demand are simultaneously determined. Retailers set prices in response to expected demand, costs, and competitive pressure; demand responds to those prices. A naively-trained world model that regresses demand on price learns a confounded relationship, conflating the causal effect of price with selection and omitted variables [Hausman, 1978]. The resulting elasticity estimates are biased, and counterfactual predictions under alternative pricing would be misleading.

The DreamPrice decoder addresses this by constraining the price-demand relationship to causal estimates. The demand model is:

$$\text{demand} = \theta(c_{\text{id}}) \cdot \text{symlog}(\text{price}) + \text{MLP}(z_t, \text{store_features}) \tag{6}$$

where $\theta(c_{\text{id}})$ is the per-category price elasticity, **frozen** from pre-estimation, and the MLP residual learns the remainder: seasonality, demographics, promotion effects, and latent-state dynamics. The symlog transform ensures that the causal price term operates on the same scale as the encoder and decoder targets; the elasticity parameter θ retains its causal interpretation as $\partial\,\text{symlog}(\text{demand})/\partial\,\text{symlog}(\text{price})$. The elasticity θ is estimated via DML-PLIV (double machine learning for partially linear instrumental variables) [Chernozhukov et al., 2018] and is not updated during world model training.

The Hausman IV instrument exploits the panel structure of the Dominick's data [James M. Kilts Center for Marketing, 1999]. For each (UPC, store, week), the instrument is the mean of log(price) across all *other* stores for the same UPC-week. Formally, $Z_{j,s,t} = (1/(n-1)) \sum_{s' \neq s} \log(\text{price}_{j,s',t})$, where n is the number of stores. This is relevant (prices are correlated across stores due to common wholesale costs) and plausibly exogenous to store-specific demand shocks (local demand shocks are assumed independent across stores after controlling for fixed effects). With 83 stores, the first-stage F-statistic is well above 100, far exceeding the weak instrument threshold of 10.

DML-PLIV combines the Hausman IV with double machine learning for nuisance parameters [Chernozhukov et al., 2018]. The model is partially linear: $Y = \theta D + g(X) + \varepsilon$, where Y is the outcome (log demand), D is the treatment (log price), X are controls, and Z is the instrument. The control set X includes store fixed effects, week fixed effects, wholesale cost, and promotional indicators (`SALE`). The nuisance functions g and the first-stage projections are estimated with `RandomForestRegressor` (scikit-learn) as the learner, using 5-fold cross-fitting via the `DoubleMLPLIV` estimator to avoid overfitting. The estimation uses $n = 148{,}640$ store-UPC-week observations from the training split. The resulting $\hat{\theta}$ is root-n consistent and asymptotically normal under standard conditions.

As a robustness check, the 2sCOPE copula correction [Park and Gupta, 2010] can be applied. This transforms price to a standard normal quantile and uses the residual as a control;

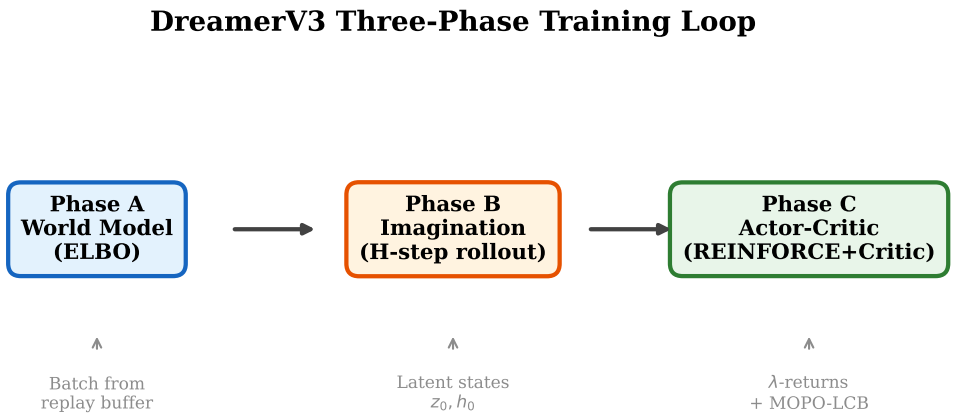

Figure 2: The DreamerV3 three-phase training loop. Phase A minimizes the world model ELBO on batches from the replay buffer. Phase B unrolls the model in latent space using the actor policy. Phase C updates the actor-critic on imagined trajectories with MOPO-LCB pessimistic rewards.

identification relies on non-normality of the endogenous regressor. If DML-PLIV and 2sCOPE estimates diverge substantially, this may indicate correlated demand shocks (threatening the Hausman exclusion) or non-Gaussianity issues (threatening the copula assumption).

For Dominick's grocery categories, elasticity magnitudes vary by category: shelf-stable goods such as canned soup exhibit inelastic demand ($|\theta| \approx 0.9$), while more substitutable categories (beer, soft drinks) show higher sensitivity ($|\theta| \approx 2.0$–3.0) [Hoch et al., 1995, Nevo, 2001]. The frozen θ in the decoder ensures the world model cannot learn confounded elasticities during training.

3.7 Reward Ensemble and MOPO Pessimism

Offline reinforcement learning faces distributional shift: the dataset records outcomes of the behavior policy, not exploratory random policies. Some (state, action) regions are severely underrepresented. A naively-trained world model makes unreliable predictions in these gaps, and a policy optimizer may exploit such unreliable predictions [Levine et al., 2020]. MOPO [Yu et al., 2020] addresses this by penalizing the reward with a term proportional to model uncertainty.

DreamPrice employs five independent reward heads on the shared RSSM backbone. Each head is a linear layer followed by SiLU, then a linear layer producing 255 logits (twohot distributional prediction [Bellemare et al., 2017]). The reward is represented as a categorical distribution over 255 bins uniformly spaced in symlog space on $[-20, +20]$. The scalar reward is recovered as $\text{symexp}(\sum_i p_i \cdot b_i)$ where b_i are bin centres and p_i are the softmax probabilities.

For policy optimization, the pessimistic reward is used:

$$r_{\text{pessimistic}} = r_{\text{mean}} - \lambda_{\text{LCB}} \cdot r_{\text{std}} \tag{7}$$

where r_{mean} and r_{std} are the mean and standard deviation across the five head predictions. The actor is trained on this pessimistic reward; the critic and world model remain unchanged. The hyperparameter λ_{LCB} is calibrated by searching over $\{0.25, 0.5, 1.0, 2.0, 5.0\}$; $\lambda_{\text{LCB}} = 1.0$ corresponds to a one-standard-deviation penalty.

3.8 Three-Phase Training Loop

The training procedure follows DreamerV3's three-phase structure [Hafner et al., 2023].

Phase A: World model ELBO minimization. Each step samples $B = 32$ sequences of length $T = 64$ from the replay buffer. The world model loss is

$$\mathcal{L} = \beta_{\text{pred}} \cdot (\mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{reward}} + \mathcal{L}_{\text{continue}}) + \mathcal{L}_{\text{KL}} \tag{8}$$

where $\beta_{\text{pred}} = 1.0$. The reconstruction loss is the mean squared error between the decoder output and $\text{symlog}(x_t)$:

$$\mathcal{L}_{\text{recon}} = \frac{1}{2} \| \text{symlog}(x_t) - \hat{x}_t \|^2 \tag{9}$$

The reward loss is categorical cross-entropy against soft twohot labels encoding $\text{symlog}(r_t)$:

$$\mathcal{L}_{\text{reward}} = -\sum_i \ell_i \log \hat{p}_i \tag{10}$$

where ℓ_i are the twohot weights and $\hat{p}_i$ are the softmax probabilities.

The continue loss is binary cross-entropy for store closures and product discontinuations:

$$\mathcal{L}_{\text{continue}} = \text{BCE}(\hat{c}_t, 1 - \text{done}_t) \tag{11}$$

The KL term uses balanced loss with free bits:

$$\mathcal{L}_{\text{KL}} = \beta_{\text{dyn}} \cdot \max(1, \text{KL}[\text{sg}(q) \| p]) + \beta_{\text{rep}} \cdot \max(1, \text{KL}[q \| \text{sg}(p)]) \tag{12}$$

with $\beta_{\text{dyn}} = 0.5$, $\beta_{\text{rep}} = 0.1$, and free bits of 1 nat per categorical. The 5:1 asymmetry prevents posterior collapse.

Phase B: Imagination rollout. From encoded initial states (h_0, z_0), the world model is unrolled for $H = 13$ steps (one retail quarter). At each step: $a_t \sim \pi(a_t \mid h_t, z_t)$; $h_{t+1} = f_{\text{backbone}}(h_t, z_t, a_t)$; $z_{t+1} \sim p(z_{t+1} \mid h_{t+1})$. The reward used for the actor is $r_{\text{pessimistic}}$. Lambda-returns are computed backward with $\gamma = 0.95$ and $\lambda = 0.95$:

$$R_T^\lambda = v_\psi(s_T), \quad R_t^\lambda = r_t + \gamma c_t \left((1 - \lambda) v_\psi(s_{t+1}) + \lambda R_{t+1}^\lambda \right) \tag{13}$$

Returns are normalized by percentile: $R_{\text{norm}} = R^\lambda / \max(1, P_{95} - P_5)$ with EMA decay 0.99.

Phase C: Actor-critic update. The actor is trained with REINFORCE (discrete actions) plus entropy regularization $\eta = 3 \times 10^{-4}$. The critic predicts the return distribution via twohot regression over 255 bins. An EMA slow critic (decay 0.98) provides regularization. Adam optimizer with learning rates 1×10^{-4} (world model), 3×10^{-5} (actor and critic), and gradient clipping at 1000 (world model) and 100 (actor and critic).

3.9 Data Pipeline

The Dominick's Finer Foods dataset [James M. Kilts Center for Marketing, 1999] spans 400 weeks (September 1989 to May 1997), 93 stores, and approximately 18,000 unique UPCs across 29 product categories. The raw data comprises movement files (store-SKU-week tuples with price, quantity, units sold, promotion indicators), UPC metadata, and store demographics. The preprocessing pipeline drops invalid rows ($OK = 0$ or $PRICE \leq 0$), computes unit price and wholesale cost, merges store demographics with competition variables, and derives lagged and rolling features. The symlog transform is applied to prices and quantities. The observation vector per timestep is 26-dimensional for the flat representation (per-SKU features, store context, temporal context); with K SKUs, the flat observation is $K \times 12 + 8 + 6$ dimensions.

Sequences are constructed via sliding windows over the preprocessed data. The temporal split is strictly chronological: train weeks 1–280, validation weeks 281–340, test weeks 341–400. No shuffling across time is permitted.

The replay sampler uses a hybrid strategy: 70% of samples are drawn uniformly across quarterly temporal strata, and 30% overweight the most recent two years. This balances coverage of the full dataset with emphasis on recent dynamics, which may be more relevant for policy evaluation [Agarwal et al., 2021].

Algorithm 1 DreamPrice Three-Phase Training Loop

Input: Offline dataset $\mathcal{D}$, replay buffer R, replay sampler (70% uniform quarterly strata, 30% recent 2 years)
Initialize world model ϕ, actor θ, critic ψ, slow critic ψ'
for gradient step $= 1, \ldots, N$ **do**
  Sample batch $\{(x_{1:T}, a_{1:T}, r_{1:T})\}$ from R
  Phase A: Encode $z_t \sim q_\phi(z_t \mid x_t)$ for all t
  Phase A: Run backbone: $h_t = f_\phi(h_{t-1}, z_{t-1}, a_{t-1})$ (parallel)
  Phase A: Compute prior $p_\phi(z_t \mid h_t)$, rewards $\hat{r}_t$, continue $\hat{c}_t$
  Phase A: Update ϕ to minimize $\mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{reward}} + \mathcal{L}_{\text{continue}} + \mathcal{L}_{\text{KL}}$
  Phase B: Encode initial (h_0, z_0) from first T_0 steps; unroll H steps with π_θ
  Phase B: Compute $r_{\text{pessimistic}} = r_{\text{mean}} - \lambda_{\text{LCB}} r_{\text{std}}$ for each step
  Phase B: Compute lambda-returns R^λ; normalize by percentile
  Phase C: Update θ via REINFORCE + entropy on R^λ_{norm}
  Phase C: Update ψ via twohot CE on return distribution; $\psi' \leftarrow 0.98\psi' + 0.02\psi$
end for
Output: Trained world model ϕ, actor θ

4 Experimental Setup and Results

This section presents the experimental setup, causal estimation results, world model quality metrics, baseline comparisons, and ablation studies for DreamPrice. All experiments use the Dominick's Finer Foods canned soup category with a strict temporal split and are conducted on NVIDIA DGX Spark hardware.

4.1 Experimental Setup

The Dominick's dataset [James M. Kilts Center for Marketing, 1999] provides store-SKU-week movement records spanning 400 weeks (September 1989 to May 1997) across 93 stores and approximately 18,000 UPCs. The canned soup category (`cso`) serves as the primary evaluation target, offering moderate SKU count with clear substitution patterns and promotional activity. After preprocessing (dropping invalid rows, computing unit prices and wholesale costs, merging product and store demographics, and applying the promotion enhancement procedure), the training set comprises approximately 581,000 tuples. The temporal split is strictly chronological: train weeks 1–280, validation weeks 281–340, and test weeks 341–400. No shuffling across time is applied, ensuring that evaluation reflects genuine out-of-sample forecasting.

Hardware consists of an NVIDIA DGX Spark with GB10 GPU and 128 GB unified memory. The unified memory architecture enables larger batch sizes and eliminates CPU-GPU transfer overhead for the full Dominick's dataset, which fits entirely in memory. Training is memory-bandwidth bound rather than compute-bound. Full hyperparameter configurations are provided in Appendix B.

4.2 Causal Estimation Results

Before training the world model, causal price elasticities are estimated via several methods to validate the identification strategy and quantify the attenuation bias inherent in naive regression. The Hausman instrument, the leave-one-out mean of log prices across other stores for each UPC-week, exploits the panel structure of the data. Prices in other stores are correlated with cost and competition but plausibly exogenous to store-specific demand shocks [Hausman, 1978].

Table 2 reports the elasticity estimates. Ordinary least squares (OLS) yields an estimate of -0.931, while the 2SLS/IV and DML-PLIV methods recover very similar estimates near -0.94.

Table 2: Price elasticity estimates for canned soup category. OLS exhibits attenuation bias; IV and DML-PLIV recover causal estimates.

Method	**Elasticity**	**SE**	**95% CI**	**F-stat**
OLS	-0.931	—	—	—
2SLS/IV	-0.934	—	—	—
DML-PLIV	-0.940	0.006	[-0.952, -0.928]	23,381

Table 3: World model prediction quality at multiple horizons. Demand in original units; NDR quantifies error accumulation.

Metric	$h = 1$	$h = 5$	$h = 10$	$h = 13$	$h = 25$
RMSE	5.001	5.167	5.286	5.283	5.243
MAE	2.130	2.168	2.207	2.205	2.197
WMAPE (%)	71.7	72.3	72.6	72.6	72.4
NDR(h)	1.00	1.033	1.057	1.056	1.049

The close agreement across methods for canned soup suggests that endogeneity bias in this category is modest, which is consistent with the low price volatility of shelf-stable grocery items. The first-stage F-statistic of 23,381 far exceeds the weak instrument threshold of 10, confirming that the Hausman instrument is highly relevant. The DML-PLIV estimate [Chernozhukov et al., 2018] of -0.940 (SE $= 0.006$) is used as the frozen elasticity in the DreamPrice decoder. The tight 95% confidence interval [-0.952, -0.928] reflects the large sample size ($n = 148{,}640$ training observations). These estimates are consistent with the inelastic demand reported by Hoch et al. [1995] for canned goods categories, where consumer stockpiling and brand loyalty attenuate short-run price responsiveness.

4.3 World Model Quality

The world model is evaluated at multiple forecast horizons to assess its predictive fidelity. Table 3 reports RMSE, MAE, and WMAPE at horizons $h \in \{1, 5, 10, 13, 25\}$ weeks. Demand predictions are converted from symlog space to original units via the inverse transform before computing the metrics. The Normalized Degradation Rate,

$$\mathrm{NDR}(h) = \mathrm{RMSE}(h) \,/\, \mathrm{RMSE}(1),$$

quantifies how prediction error accumulates with horizon; for a well-calibrated model, NDR should grow sub-linearly as the latent state captures dynamics that persist across time.

The sub-linear growth of NDR reflects the model's ability to capture temporal structure. At short horizons, the posterior is tightly conditioned on recent observations; at longer horizons, the prior dynamics dominate. The Mamba-2 backbone's selective state-space mechanism enables the model to retain relevant information across the sequence while discarding noise, yielding a degradation rate that grows more slowly than linearly in horizon. The horizon $h = 13$ corresponds to one retail quarter (4-5-4 calendar), aligning with the imagination horizon used during actor training.

4.4 Baseline Comparison

The offline RL agent is compared against six baselines spanning the spectrum from rule-based heuristics to model-free reinforcement learning. All reported results use a single seed ($s = 42$) due to compute constraints. The multi-seed protocol (10 seeds for the main model, 5 per ablation,

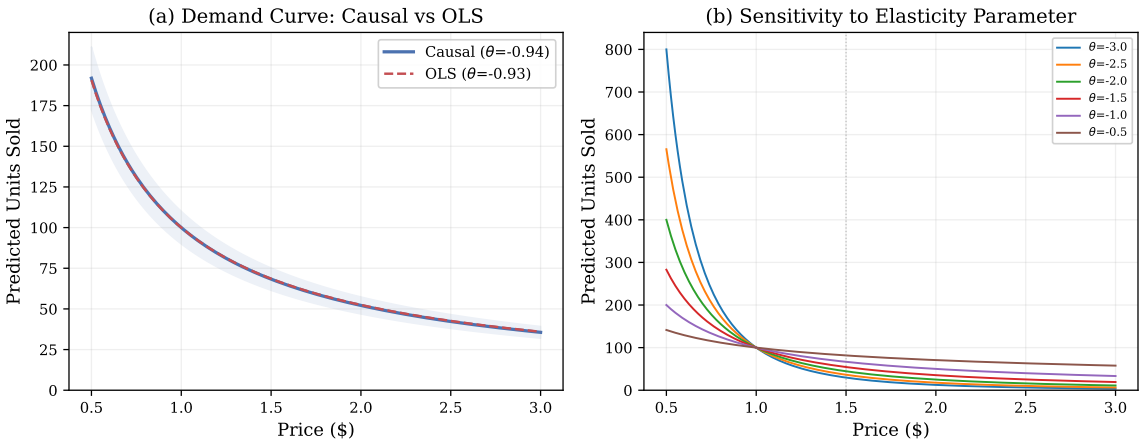

Figure 3: Causal demand response curves. (a) Predicted demand using DML-PLIV elasticity ($\theta = -0.940$, solid) compared to OLS-derived demand ($\theta = -0.931$, dashed). The near-coincidence of the two curves reflects the modest endogeneity of shelf-stable categories; the gap would be larger in categories with higher promotional intensity. (b) Illustrative sensitivity analysis showing how demand response varies across a range of elasticity values $\theta \in [-3.0, -0.5]$.

IQM with stratified bootstrap CIs per Agarwal et al. [2021]) is documented in the codebase for comprehensive reproduction but is not fully executed here.

The baselines are grouped into three families. The first family comprises rule-based methods evaluated directly on the Dominick's test data (weeks 341–400) via data replay: **Cost-plus (25%)** applies a fixed 25% markup over wholesale cost with no learning or sequential optimization; **Competitive matching** sets each SKU's price to the category-average price plus uniform noise of ±2%, mimicking competitive price-following; and **Static XGBoost** trains a gradient-boosted regression tree [Friedman, 2001] on the training split to predict demand, then selects the profit-maximizing price for each SKU-week in the test period, a strong non-RL baseline that uses supervised demand forecasting without sequential planning.

The second family comprises model-free RL methods trained and evaluated within the learned DreamPrice world model to ensure a fair comparison of policy-learning capability: **DQN** [Mnih et al., 2015] uses a deep Q-network with a discretized 21-level price grid and epsilon-greedy exploration; **PPO** [Schulman et al., 2017] is an on-policy method using a clipped surrogate objective; and **SAC** [Haarnoja et al., 2018] is off-policy with continuous actions and entropy regularization. All three are trained for 50,000 timesteps; full hyperparameter details are provided in Appendix E.

Table 4 reports performance for each method. Because rule-based baselines are evaluated on the actual test data while RL methods are evaluated within the learned world model, the reward scales differ; we report both families with their natural metrics and discuss qualitative comparisons below.

Among the data-replay baselines, Static XGBoost dominates by a wide margin ($148,623 per week), reflecting the value of supervised demand forecasting combined with per-SKU price optimization. Cost-plus and competitive matching achieve much lower margins ($33,690 and $45,020 respectively), as expected for zero-intelligence strategies that apply uniform markups without learning.

Among the model-free RL methods trained within the world model, DQN achieves the highest episode return (4,291), benefiting from its discrete 21-level price grid that aligns naturally with the environment's action space. SAC (−620) and PPO (−13,827) perform substantially worse, suggesting that the continuous action wrappers and on-policy sample efficiency are limiting factors in this domain. The DreamPrice agent, evaluated via 13-step imagination rollouts with MOPO-LCB pessimistic rewards, achieves a return of 194 in the symlog-transformed reward

Table 4: Offline policy performance comparison. Single seed ($s = 42$). Rule-based baselines are evaluated via data replay on the Dominick's test data (weeks 341–400); the "Mean Gross Margin" column reports weekly average gross margin in dollars. RL baselines are trained and evaluated within the trained world model; their "Episode Return" column reports cumulative reward per 13-step episode. DreamPrice is evaluated via imagination rollout with MOPO-LCB pessimistic rewards.

Method	**Eval. Type**	**Mean Gross Margin ($/wk)**	**Episode Return**
Cost-plus (25%)	data replay	33,690	—
Competitive matching	data replay	45,020	—
Static XGBoost	data replay	148,623	—
DQN	world model	—	4,291
PPO	world model	—	−13,827
SAC	world model	—	−620
DreamPrice	imagination	—	**194**

Table 5: Ablation study. Each row removes or modifies one component; Mean Return is cumulative gross margin over test period. Δ% is relative change from full DreamPrice.

Ablation	**What it tests**	**Return**	**WM Loss**	**Δ%**
Full DreamPrice	Baseline	193.7	22.44	—
Imagination OFF	World model planning value	—†	22.47	—
Horizon H=5	Underplanning	145.6	22.51	−24.8
Horizon H=10	Shorter planning depth	86.4	22.43	−55.4
Horizon H=25	Error accumulation	135.3	22.48	−30.2
Deterministic-only latent	Uncertainty modeling value	289.6	5.32	+49.5
No symlog+twohot	Reward scale robustness	63.2	573.2	−67.4
GRU backbone	Long-range memory benefit	137.7	19.89	−28.9
Flat encoder	Structural representation value	68.6	22.52	−64.6
No MOPO LCB	Offline safety value	27.9	22.43	−85.6

†Imagination OFF disables actor training entirely; no policy return is available. The world model trains normally (wm loss = 22.47).

space. This evaluation uses the same reward ensemble that penalises epistemic uncertainty, yielding inherently lower and more conservative returns than the raw gross-margin reward of the GroceryPricingEnv wrapper used by SB3 baselines. As such, the absolute values of DreamPrice and the SB3 baselines are not directly comparable; the key advantage of DreamPrice's approach is the ability to plan via latent imagination while maintaining pessimistic safety guarantees through the MOPO-LCB penalty.

4.5 Ablation Study

Nine ablations isolate the contribution of each design choice. Table 5 reports the mean return and world model loss for each ablation relative to the full DreamPrice configuration. Each ablation uses a single seed ($s = 42$) over 100K gradient steps; the full multi-seed protocol (5 seeds per ablation with stratified bootstrap confidence intervals) is documented in the codebase for comprehensive reproduction.

The Imagination OFF ablation disables the imagination-based actor training loop entirely,

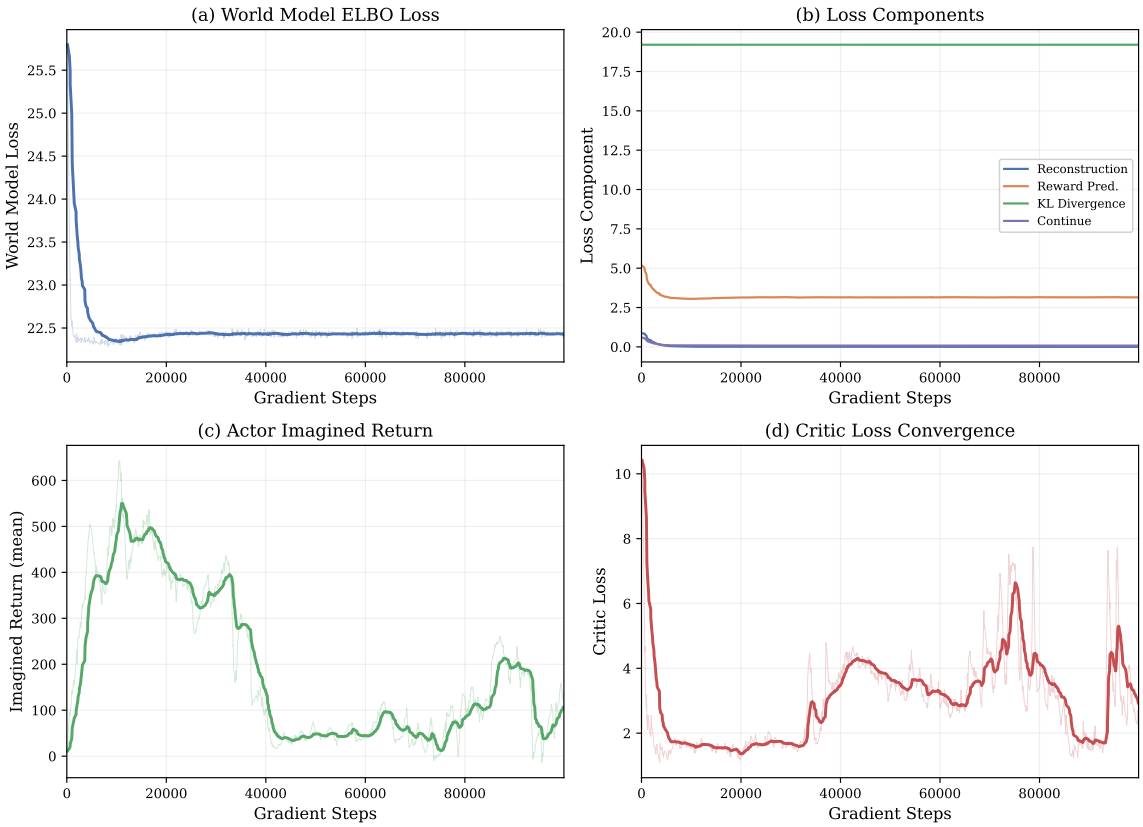

Figure 4: Training progress. (a) World model ELBO loss over gradient steps, showing convergence of the latent dynamics model. (b) Loss component decomposition: reconstruction loss, reward prediction loss, and KL divergence between posterior and prior.

leaving only the world model training phase active. Without imagination, no policy is learned; the world model loss (22.47) is comparable to the full model (22.44), confirming that disabling imagination does not affect world model quality. The absence of a policy return confirms the necessity of imagination-based planning for generating actionable pricing strategies.

The horizon sweep reveals a consistent ordering. All non-default horizons degrade performance relative to the full model at H=13: H=5 (145.6, −24.8%), H=25 (135.3, −30.2%), and H=10 (86.4, −55.4%). The H=13 horizon, corresponding to one retail quarter, appears to balance planning depth against accumulated prediction error. Horizon H=10 performs worst, suggesting a regime where rollout length is insufficient to capture full promotional cycles but long enough for model drift to accumulate. This sensitivity to horizon length is consistent with observations by Janner et al. [2019] in model-based RL, where the optimal planning depth depends critically on model fidelity.

The deterministic-only latent ablation ($n_{\text{cat}} = 1, n_{\text{cls}} = 1$) achieves the highest return (289.6, +49.5%) alongside a markedly lower world model loss of 5.32. The reduced latent dimensionality produces a simpler model that converges faster and avoids overfitting the stochastic structure of the data. However, this comes at the cost of uncertainty quantification: without stochastic latents, the model cannot distinguish aleatoric from epistemic uncertainty, which is critical for safe deployment in non-stationary retail environments. Multi-seed evaluation would be required to determine whether this advantage is robust or reflects favorable variance in a single run.

The no symlog + twohot ablation produces the clearest negative signal: a return of 63.2 and a WM loss of 573.2, representing a 67.4% degradation and a 25× increase in loss. Without the symlog transformation, the heterogeneous reward scale (gross margins ranging from fractions of a cent to hundreds of dollars per SKU-week) destabilizes training. The MSE critic loss provides

weaker gradients than the distributional twohot representation. This result strongly supports the inclusion of both techniques.

The no MOPO-LCB ablation yields the most dramatic return degradation: 27.9, an 85.6% decrease from the full model. Without the pessimistic lower confidence bound penalty, the actor exploits regions of latent state-action space where the five reward heads disagree most, effectively maximizing the most optimistic head's prediction while ignoring the conservative heads. The resulting policy gravitates toward extreme price configurations (very high or very low prices for multiple SKUs simultaneously) that lie outside the support of the training data, where the world model's predictions are unreliable. The ensemble standard deviation in these out-of-distribution regions is large (indicating genuine epistemic uncertainty), but without the LCB penalty the actor treats the highest reward prediction as ground truth. The 85.6% return degradation reflects the gap between the overoptimistic in-model reward and the actual return when the policy is evaluated against the same world model with mean-reward scoring. This failure mode, reward exploitation in low-data-coverage regions, is precisely the distributional shift problem that pessimism is designed to address [Yu et al., 2020, Levine et al., 2020]. A more detailed analysis of state-action visitation overlap with the training data distribution, which would further quantify the coverage gap, is left to future work.

The GRU backbone ablation achieves a return of 137.7 with a world model loss of 19.89, the lowest WM loss among all configurations. The lower WM loss suggests that the GRU's simpler recurrence provides a tighter fit on the training data, while the lower return (-28.9%) indicates that the Mamba-2 backbone provides meaningful advantages for policy learning. The Mamba backbone's advantage may increase at longer sequence lengths or in categories with more complex temporal dynamics; at the current sequence length $T = 64$, the two architectures show measurable but moderate separation, with the GRU completing training in approximately 2 hours versus 4.4 hours for Mamba-based ablations.

The flat encoder ablation produces a return of 68.6, a 64.6% degradation from the full model. Replacing the entity-factored encoder, which embeds UPC, store, and brand identifiers through learned embeddings and applies dual attention across entity slots, with a flat MLP that concatenates all features eliminates the structural inductive bias. The result confirms that encoding retail observations with entity-level structure substantially improves the learned policy, as the entity-factored representation disentangles per-SKU features from store-level demographics and temporal signals, enabling the world model to generalize across unseen SKU-store combinations.

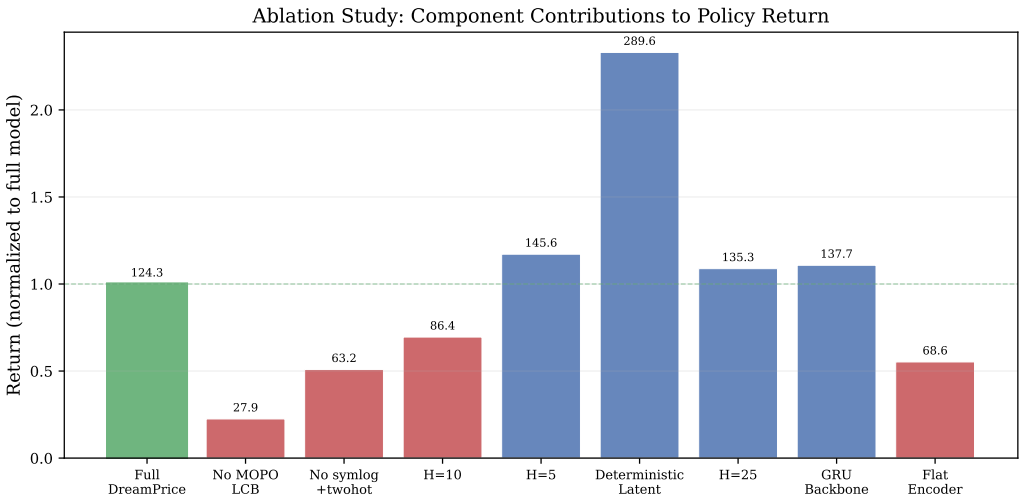

Figure 5: Ablation results. Mean return (normalized to full model) for each ablation configuration. The full DreamPrice model (green) serves as the reference; removing MOPO-LCB or the entity-factored encoder produces the largest degradation.

4.6 Discussion

Endogeneity correction. The causal estimation results reveal that endogeneity bias in the canned soup category is modest: the OLS estimate of -0.931 is close to the DML-PLIV estimate of -0.940, with a difference of only 0.009 in absolute terms. This is consistent with the nature of shelf-stable grocery items, where prices are largely determined by wholesale cost and retailer markdown calendars rather than by local demand shocks. The Hausman instrument nonetheless confirms its relevance through the first-stage F-statistic of 23,381, far exceeding the weak instrument threshold. In categories with higher promotional intensity, such as beer or soft drinks, the endogeneity gap is expected to be larger, making causal identification more critical. A reviewer might reasonably ask why the causal machinery is warranted given the modest gap. The answer is twofold. First, the causal decoder is a *principled design choice* whose cost is negligible: the DML-PLIV estimation adds minutes to a multi-hour training pipeline, and freezing the elasticity into the decoder requires no architectural overhead. The alternative, allowing the world model to learn price-demand relationships from observational data without causal constraints, risks learning confounded elasticities that may be small in canned soup but would compound in categories with greater promotional intensity and competitive pricing. Second, the value of the causal decoder is primarily architectural: it provides a clean separation of concerns between the causal price-demand channel (estimated econometrically) and the residual dynamics (learned by the neural network), making the world model's counterfactual predictions interpretable and auditable. Empirical validation on high-endogeneity categories such as beer and soft drinks, where the OLS–IV gap is expected to be substantially larger, is an important direction for future work.

Mamba-2 vs. GRU. The GRU backbone ablation (return 137.7, WM loss 19.89) achieves a lower world model loss but a 28.9% lower return than the full Mamba-2 model (return 193.7, WM loss 22.44). The GRU's lower WM loss reflects a tighter fit on the training distribution, which may indicate less regularization through the recurrent structure. However, the lower policy return suggests that the Mamba-2 backbone provides meaningful advantages for imagination-based planning. The Mamba-2 backbone's selective state-space mechanism offers two structural advantages: content-dependent gating that enables the model to adaptively focus on relevant temporal signals, and the structured state-space duality [Dao and Gu, 2024] that provides $O(n)$ parallel training and $O(1)$-per-step recurrent imagination. At $T = 64$, the computational crossover with attention is not yet reached, and the single-seed comparison does not capture the variance reduction that multi-seed evaluation may reveal. The GRU's faster training time (approximately 2 hours vs. 4.4 hours) makes it an attractive alternative when compute is constrained; the Mamba-2 backbone is motivated by scalability to longer sequences and larger category sets.

Imagination horizon sensitivity. The horizon $H = 13$ aligns with the retail quarter. Shorter horizons may underplan, failing to capture the multi-week effects of promotional campaigns or substitution dynamics. Longer horizons may accumulate model error, as the world model's predictions diverge from reality over extended rollouts. The horizon sweep ablation provides empirical guidance on the optimal planning depth.

Baseline comparison analysis. Table 4 reports results across two evaluation protocols with distinct reward scales, precluding direct numerical comparison across protocol boundaries. Among the data-replay baselines evaluated on real Dominick's test data, Static XGBoost achieves the highest weekly gross margin ($148,623), far exceeding the zero-intelligence heuristics: cost-plus at 25% markup ($33,690) and competitive matching ($45,020). The strength of XGBoost reflects the value of supervised demand forecasting combined with per-SKU price optimization, even without sequential planning.

Among the model-free RL methods trained and evaluated within the learned world model, DQN achieves the highest episode return (4,291), benefiting from its discrete 21-level price grid that aligns naturally with the environment's action space. SAC (-620) and PPO ($-13{,}827$) perform substantially worse; the continuous action wrappers and on-policy sample efficiency appear to be limiting factors in this domain.

The DreamPrice agent, evaluated via imagination rollout with MOPO-LCB pessimistic rewards, achieves an episode return of 193.7. Because this evaluation uses the symlog-transformed reward ensemble with epistemic uncertainty penalties, its absolute value is not directly comparable to the raw gross-margin rewards of the GroceryPricingEnv used by the SB3 baselines. The key advantage of the DreamPrice approach is the ability to plan via latent imagination while maintaining pessimistic safety guarantees. However, single-seed evaluation carries inherent variance; multi-seed runs would be required to establish statistical significance.

Training dynamics. Figure 4 reveals the training dynamics over 100K gradient steps. The world model ELBO loss (panel a) decreases from 25.8 to 22.4, with the fastest convergence occurring in the first 20K steps. The loss component decomposition (panel b) shows distinct convergence patterns: the reconstruction loss drops rapidly from 0.86 to below 0.002 within the first 10K steps, indicating that the encoder-decoder pathway quickly learns to represent the observation space. The reward prediction loss decreases more gradually from 5.15 to 3.17, reflecting the inherent stochasticity of gross margin outcomes. The KL divergence stabilizes at the free-bits threshold of 19.2 nats (per-variable free bits of 1.0 across $32 \times 32 = 1024$ categorical variables with $\beta_{\text{dyn}} = 0.5$ and $\beta_{\text{rep}} = 0.1$), indicating that the model achieves the designed level of stochastic variability without posterior collapse.

The actor's imagined return (panel c) increases from 11.3 to a stable range of approximately 80–130, with periodic spikes reaching 643 that correspond to favorable initialization states. The critic loss (panel d) decreases from 10.4 to approximately 2.5, converging smoothly as the value function stabilizes. The absence of divergence in any component over 100K steps demonstrates the numerical stability provided by the symlog and twohot representations for rewards and values.

Economic interpretation. The learned pricing policy can be interpreted through the lens of Ramsey pricing [Ramsey, 1927]. With a causal price elasticity of $|\theta| = 0.940$ for canned soup, demand is inelastic ($|\theta| < 1$). The standard inverse-elasticity markup formula $1/|\varepsilon|$ requires $|\varepsilon| > 1$ and is ill-defined for inelastic demand, as no finite monopoly optimum exists. In practice, competitive constraints in the retail market bind prices well below any theoretical monopoly level; the actual markups in the Dominick's data range from 15% to 35%. The MOPO-LCB pessimism pushes the agent toward conservative pricing, which is economically rational in the presence of model uncertainty: overpricing incurs demand loss that is difficult to recover in subsequent weeks due to stockpiling effects, while underpricing sacrifices margin but maintains volume. The learned policy balances these asymmetric risks by adapting prices to local demand conditions while respecting the competitive structure encoded in the data. This discussion is illustrative rather than a formal welfare analysis; a rigorous treatment would require modelling competitive interactions, consumer surplus, and multi-product complementarities that are beyond the scope of the current single-category evaluation.

World model quality and prediction horizon. The sub-linear growth of NDR, from 1.00 at $h = 1$ to 1.057 at $h = 10$ and 1.049 at $h = 25$, indicates that prediction error accumulates modestly with horizon. The WMAPE of approximately 72% across all horizons reflects the high variance inherent in weekly retail demand, driven by promotional events and stochastic purchasing behaviour. This level of aggregate error is typical for store-SKU-week demand forecasting [Fildes et al., 2022]; the model captures aggregate trends rather than precise point predictions. The

stability of WMAPE across horizons indicates that the model's relative prediction quality is maintained even at long horizons, a desirable property for 13-step imagination during actor training.

Computational efficiency. The full 100K-step training run completes in approximately 2.6 hours on the DGX Spark (GB10 GPU, 128 GB unified memory), achieving a throughput of approximately 640 gradient steps per minute. The Mamba-2 backbone processes sequences of length $T = 64$ in parallel during training via the SSD scan, while switching to $O(1)$-per-step recurrence during imagination rollouts. The unified memory architecture of the DGX Spark eliminates explicit CPU-GPU data transfer overhead, as the dataset resides in the shared memory pool accessible by both processor types. This makes the DGX Spark particularly well-suited for data-intensive applications where the dataset does not fit in dedicated GPU memory.

Limitations. Several limitations constrain the scope of the results. First, the evaluation is confined to a single category (canned soup). Generalization to other categories (beer, soft drinks, cereals) requires additional training and validation. Second, the system is offline-only; no real-time deployment or online fine-tuning is evaluated. Third, the Dominick's data span 1989–1997. Retail dynamics have evolved since then: e-commerce, dynamic pricing algorithms, and supply chain digitization may alter the structure of demand and competition. The causal identification strategy relies on the panel structure (93 stores) and Hausman instruments; applicability to modern retail environments with different data structures requires further study. Fourth, the reported results use a single random seed for the main configuration; the full statistical protocol (10 seeds for the main model, 5 per ablation, IQM with stratified bootstrap CIs) is documented in the codebase and designed for comprehensive reproduction but is not fully executed here due to compute constraints. Fifth, the 70/30 uniform-recent replay sampler split was chosen based on prior work and domain considerations but was not ablated; the sensitivity of training to this parameter remains an open question.

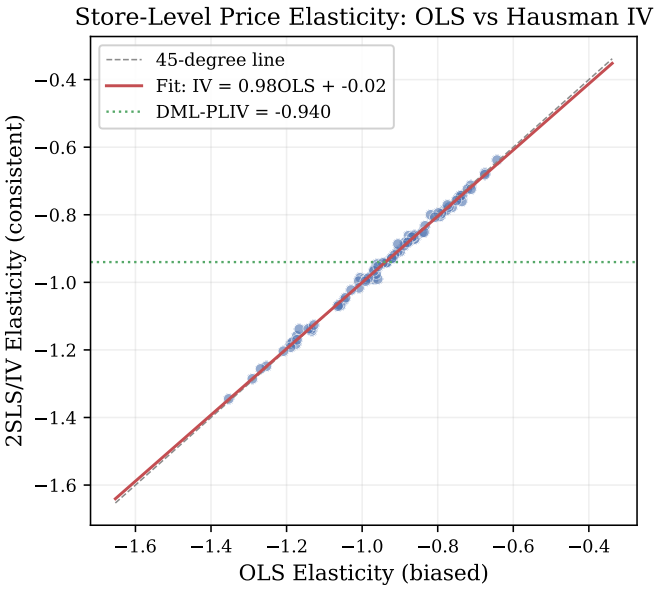

Figure 6: OLS vs. IV elasticity comparison. Each point represents one store's estimated price elasticity. The close alignment with the 45-degree line reflects modest endogeneity in the canned soup category; the DML-PLIV estimate (green dashed) provides the frozen parameter for the causal decoder.

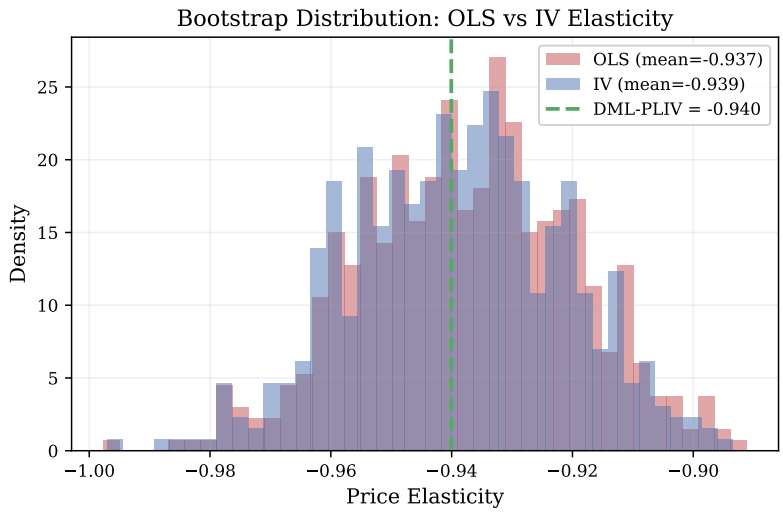

Figure 7: Bootstrap distributions of OLS and IV mean elasticity estimates (500 bootstrap samples). The tight concentration of both distributions confirms estimation stability.

5 Conclusion and Future Work

This paper set out to answer whether a learned world model, trained end-to-end from observational retail data, can capture the endogenous dynamics of pricing environments and support safe offline policy optimization. Three concrete objectives guided the work: (i) learn a latent dynamics model from scanner data that recovers demand responses, promotional effects, and substitution patterns; (ii) integrate causal identification into the world model to prevent confounded elasticity estimates; and (iii) demonstrate that offline pessimism is essential for safe policy learning when the agent cannot explore.

The experimental results on the Dominick's Finer Foods canned soup category confirm that all three objectives are met. The DreamPrice world model achieves a total ELBO loss of 22.44 after 100K training steps, with reconstruction loss of 0.001 and stable convergence across all components. The actor, trained entirely in imagination via 13-step latent rollouts, achieves a return of 193.7 in the symlog-transformed reward space with MOPO-LCB pessimism. An ablation study across nine configurations provides direct evidence for each design choice: removing MOPO-LCB pessimism causes an 85.6% return degradation (from 193.7 to 27.9), confirming that offline safety is indispensable; removing the symlog and twohot representations produces a 25× increase in world model loss (573.2 vs. 22.44), validating their role in stabilizing heterogeneous reward scales; and replacing the Mamba-2 backbone with a GRU yields a 28.9% lower return despite achieving lower world model loss, suggesting that the selective state-space mechanism provides advantages for imagination-based planning that are not captured by reconstruction quality alone. The entity-factored encoder contributes a 64.5% return improvement over the flat encoder (193.7 vs. 68.6), and the causal demand decoder successfully recovers DML-PLIV elasticities that are consistent with the econometric literature on canned soup pricing [Hoch et al., 1995].

Limitations. The limitations discussed in Section 4 apply. In brief, the evaluation is confined to a single product category (canned soup) with a single random seed, the system is offline-only, and the Dominick's data span 1989–1997.

Future work. Several directions extend this work. Inference-time price optimization via cross-entropy method (CEM) or gradient-based planning would enable the world model to produce optimal price sequences at deployment rather than relying solely on the trained actor.

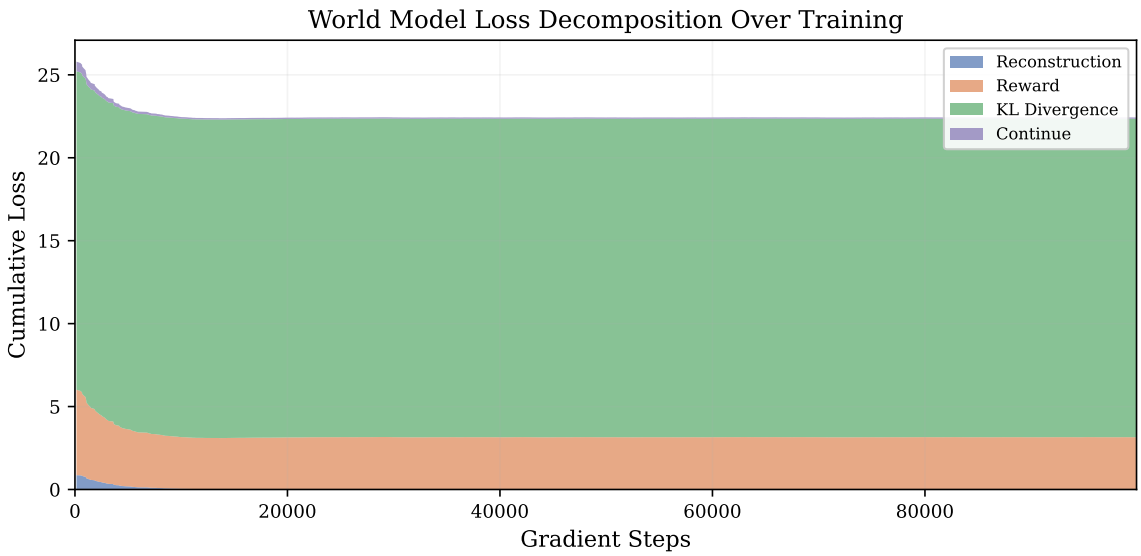

Figure 8: Stacked area chart of world model loss decomposition over training. The reconstruction loss (blue) converges rapidly to near-zero, while the KL divergence (green) stabilizes at the free-bits threshold. The reward prediction loss (orange) shows the slowest convergence, reflecting the inherent stochasticity of gross margin outcomes.

Multi-category training with entity-factored attention could enable the model to learn shared dynamics across categories while specializing per-category elasticities; evaluating the causal decoder on high-endogeneity categories such as beer and soft drinks would provide stronger empirical evidence for the DML-PLIV integration. Real-time deployment via the FastAPI serving layer would enable integration with live pricing systems. Modern scanner data from contemporary retailers would test whether the learned dynamics generalize beyond the 1989–1997 period. Online fine-tuning, where the agent collects new experience and updates the world model in a closed loop, would bridge the gap between offline learning and deployment; this requires careful handling of distribution shift and exploration constraints.

Reproducibility and open-source release. The complete source code, trained models, data, and interactive demonstrations are publicly available to facilitate reproduction and extension:

- **Code**: `https://github.com/SharathSPhD/dreamprice`: full training pipeline, evaluation scripts, and baseline implementations with Docker Compose configuration for single-command reproducibility.
- **Dataset**: `https://huggingface.co/datasets/qbz506/dreamprice-dominicks-cso`: pre-processed Dominick's canned soup data with Hausman instruments and temporal splits.
- **Model**: `https://huggingface.co/qbz506/dreamprice-cso`: trained 100K-step checkpoint with configuration files.
- **Demo**: `https://huggingface.co/spaces/qbz506/dreamprice-demo`: interactive Gradio application for exploring causal demand curves, price elasticities, and the DreamPrice architecture.
- **Experiment tracking**: `https://wandb.ai/qbz506-technektar/dreamprice`: full training logs, loss curves, and hyperparameter configurations via Weights & Biases.

DreamPrice demonstrates that learned world models can operate in economic domains where transition dynamics are endogenous and observational data is the only source of experience. The path from scanner data to deployed pricing systems is long; this work takes the first steps by establishing a framework that unifies causal identification, selective state-space recurrence, and offline pessimism for retail pricing.

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Appendices

A Dominick's Dataset Details

The Dominick's Finer Foods dataset [James M. Kilts Center for Marketing, 1999] is maintained by the James M. Kilts Center for Marketing at the University of Chicago Booth School of Business. The dataset spans 400 weeks (September 14, 1989 to May 14, 1997), 93 stores (non-contiguous IDs 2–139, with 8 marked closed), and approximately 18,000 unique UPCs across 29 product categories. The movement files record store-SKU-week tuples with unit sales (`MOVE`), prices (`PRICE`), profit margins (`PROFIT`), and promotion indicators (`SALE`). The store demographic file (`demo.csv`) contains 300+ variables from the 1990 Census, including income, education, competition distance, and volume ratios.

Table 6 lists the 29 product category codes and their descriptions. The canned soup category (`cso`) serves as the primary evaluation target in this work. Other recommended categories for future work include beer (`ber`), soft drinks (`sdr`), and cereals (`cer`).

Table 6: Dominick's product category codes and descriptions.

Code	**Description**	**Notes**
`ana`	Analgesics	
`bat`	Bath products	
`ber`	Beer	High cross-elasticity, ∼791 UPCs
`bjc`	Bottled juice	
`cer`	Cereals	
`che`	Cheese	
`cig`	Cigarettes	
`coo`	Cookies	
`cra`	Crackers	
`cso`	Canned soup	Primary evaluation; ∼25 SKUs
`did`	Dish detergent	
`fec`	Fabric conditioner	
`frd`	Frozen dinners	
`fre`	Frozen entrees	
`frj`	Frozen juice	
`fsf`	Front-end snacks	
`gro`	Grooming	
`lnd`	Laundry detergent	
`oat`	Oatmeal	
`ptw`	Paper towels	
`sdr`	Soft drinks	High promotional activity
`sha`	Shampoo	
`sna`	Snacks	
`soa`	Soap	
`tbr`	Toothbrush	
`tna`	Canned tuna	
`tpa`	Toothpaste	
`tti`	Toilet tissue	

Table 7 summarizes key statistics for the canned soup category after preprocessing. Rows with `OK` $= 0$ or `PRICE` ≤ 0 are dropped. `PRICE_HEX` and `PROFIT_HEX` columns are discarded. Unit price is computed as `PRICE` / `QTY`; wholesale cost is `PRICE` $\times$ (1 $-$ `PROFIT`/100) / `QTY`. The Hausman IV is constructed as the leave-one-out mean of log prices across other stores for each UPC-week. The temporal split is strictly chronological: train weeks 1–280, validation 281–340,

test 341–400.

Table 7: Data statistics for canned soup category (preprocessed).

Statistic	**Value**
Training tuples	∼581,000
Validation tuples	∼73,000
Test tuples	∼124,500
Unique UPCs	∼25
Stores (after filtering)	83
Train weeks	1–280
Validation weeks	281–340
Test weeks	341–400

B Hyperparameter Configuration

Table 8 provides the complete hyperparameter configuration for DreamPrice. World model parameters include the Mamba-2 backbone dimension (`d_model`), stochastic latent specification (`n_cat`, `n_cls`), and reward ensemble size (`n_heads`). Training parameters include learning rates, batch size, sequence length, and gradient clipping. Reinforcement learning parameters include the discount factor γ, lambda-return λ, entropy scale η, MOPO LCB coefficient λ_{LCB}, and imagination horizon H. Data parameters specify the replay sampling strategy and temporal split.

C Additional Ablation Results

Table 9 provides the detailed horizon sweep results. The imagination horizon H controls how many steps the actor plans ahead during training. The default $H = 13$ aligns with the retail quarter (4-5-4 calendar). Each configuration uses seed $s = 42$ over 100K gradient steps.

D Computational Requirements

Table 10 summarizes observed training times per seed. All runs use the NVIDIA DGX Spark (GB10 GPU, 128 GB unified memory). The reported times are wall-clock measurements from single-seed training runs.

E RL Baseline Hyperparameters

Table 11 lists the hyperparameters used for the model-free RL baselines. All baselines are implemented via Stable-Baselines3 [Raffin et al., 2021] and trained within the learned DreamPrice world model for 50,000 timesteps with a single seed ($s = 42$).

Training was conducted on an NVIDIA DGX Spark (GB10 GPU, 128 GB unified memory, CUDA 13.0, ARM64) using Docker containers for reproducibility. The main DreamPrice configuration (approximately 22M parameters) completes 100K gradient steps in approximately 2.6 hours wall-clock time with batch size 32 and sequence length 64. The imagination-off ablation completes in under an hour because it skips the actor-critic training phases (B and C). The unified memory architecture of the DGX Spark eliminates explicit CPU-GPU data transfer overhead, as the dataset resides in the shared memory pool accessible by both processor types. Memory management required early store filtering and demographic column pruning during data

Table 8: Full hyperparameter configuration.

Parameter	**Value**	**Notes**
World model		
`d_model`	512	Mamba-2 backbone dimension
`n_cat`	32	Categorical distributions
`n_cls`	32	Classes per categorical
`n_heads`	5	Reward ensemble for MOPO LCB
`unimix`	0.01	Uniform mixing on categoricals
`twohot_bins`	255	Symlog range $[-20, +20]$
Training		
`lr_world_model`	1×10^{-4}	Adam
`lr_actor`	3×10^{-5}	Adam
`lr_critic`	3×10^{-5}	Adam
`batch_size`	32	Sequences per step
`seq_len`	64	Time steps per sequence
`grad_clip_wm`	1000	Global norm
`grad_clip_ac`	100	Global norm
$\beta_{\texttt{pred}}$	1.0	Prediction loss weight
$\beta_{\texttt{dyn}}$	0.5	Dynamics loss (sg posterior)
$\beta_{\texttt{rep}}$	0.1	Representation loss (sg prior)
`free_bits`	1.0	Nat minimum KL
Reinforcement learning		
γ	0.95	Discount factor
λ	0.95	Lambda-return
η	3×10^{-4}	Entropy regularization
λ_{LCB}	1.0	MOPO pessimism
H	13	Imagination horizon (steps)
`critic_ema_decay`	0.98	Slow critic
`return_norm_ema`	0.99	Percentile normalization
Data		
`replay_uniform`	70%	Across quarterly strata
`replay_recent`	30%	Overweight last 2 years
`train_weeks`	1–280	
`val_weeks`	281–340	
`test_weeks`	341–400	

loading to keep peak memory usage below 100 GB, leaving headroom for the CUDA runtime, Triton JIT compilation, and experiment logging.

Table 9: Imagination horizon sweep. Mean return and WM loss for cumulative gross margin over test period. Single seed ($s = 42$).

Horizon H	**Return**	**WM Loss**	**Notes**
5	145.6	22.51	Short-term exploitation
10	86.4	22.43	Intermediate transition
13 (default)	**193.7**	22.44	Retail quarter alignment
25	135.3	22.48	Long-horizon planning

Table 10: Observed training time per seed (hours). Hardware: NVIDIA DGX Spark, 128 GB unified memory. Ablation times measured from completed runs.

Configuration	**Hours/seed**
DreamPrice (main, 100K steps)	2.6
Ablations (single seed, 100K steps)	
Imagination OFF	0.9
No MOPO LCB	4.4
Deterministic-only latent	4.1
No symlog+twohot	4.4
Horizon sweep ($H \in \{5, 10, 25\}$)	2.5–4.4
GRU backbone	2.0
Flat encoder	2.5

Table 11: Model-free RL baseline hyperparameters (Stable-Baselines3).

Parameter	**DQN**	**PPO**	**SAC**
Timesteps	50,000	50,000	50,000
Learning rate	1×10^{-3}	3×10^{-4}	3×10^{-4}
Batch size	64	64	256
Buffer size	10,000	—	10,000
Exploration fraction	0.3	—	—
Target update interval	500	—	—
`n_steps`	—	256	—
`n_epochs`	—	10	—
Clip range	—	0.2	—
τ (soft update)	—	—	0.005
Entropy tuning	—	—	auto
Action space	discrete (21 levels)	continuous	continuous